A Heterogeneous Model for Simulating Startup Growth and Failure: Literature Review

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1. Introduction. Venture capitalists (VCs) allocate enormous resources under uncertainty. In 2024, over $368 billion was invested globally by VCs in startups [6], despite empirical studies estimating that roughly 75% of venture-backed startups fail [10]. VCs must make decisions on whether to invest in a startup based on limited information about the startup's early trajectory of success indicators, such as revenue growth, customer acquisition, and cash reserves. Inspired by the difficulty and economic importance of this problem, we are interested in studying methods of simulating startup outcomes in a VC context. In particular, we look to develop a stochastic simulator that models startup growth and failure to produce a range of potential company trajectories based on estimated company-specific parameters.

2. Existing Methods. At early funding rounds, such as Pre-Seed, Seed, and Series A, VC decisions are often made entirely based on experience and intuition, without models. Models are introduced in later rounds, where historical data is more plentiful to fit a simulator's parameters. We aim to eliminate some of the stratification in modeling startup valuations by creating a unified simulator applicable to all funding rounds. The added uncertainty in early rounds will be reflected as increased variability (uncertainty) in our simulator's output, such as high variance in valuation, growth, and customer acquisition distributions.

Methods such as Customer-Based Corporate Valuation (CBCV) are common past Series C [9]. It is a relatively simple model that splits target valuation into steady-state value (current recurring value, no growth) and value-creating investments (pure growth). This model requires many relevant statistics for both the startup and its competitors, such as the mean customer lifetime value (LTV), customer acquisition cost (CAC), and market size. It models revenue growth by proxy, directly estimating customer growth and churn. In a similar fashion, we aim to model performance based on customer statistics, not directly on revenue or profit.

After Series D, a Stochastic Discounted Cash Flow (SDCF) model [7], a stochastic modification to the standard DCF model, can be incorporated with Monte Carlo simulations to generate a full probability distribution of possible valuation figures, modeling uncertainty instead of just producing a point estimate. Additionally, a method known as Real Options Valuation (ROV) [8] has shown success, which models investments in a particular round as a call option on the subsequent round, giving the VC the right–but not the obligation–to invest further in that subsequent round. This idea can be used to construct a Binomial Option Pricing Model (BOPM), which gives decision trees to map out discrete funding stages and calculate neutral probabilities of the startup's value moving up or down at each node.

3. Revenue Growth Models. Revenue growth is often modeled by combining the predictable patterns of consumer adoption with the random fluctuations found in real-world markets.

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In his 1969 paper, Frank Bass modeled the spread of new products using the following differential equation [1]:

$$f(t) = \frac{dF(t)}{dt} = (p + qF(t))(1 - F(t))$$

In this model, p is the coefficient of innovation, which represents external marketing, and q is the coefficient of imitation, which represents internal word-of-mouth. This equation can be solved to find $F(t)$, the fraction of the market reached, and the resulting revenue:

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}} \quad R_t = M \cdot a \cdot F(t)$$

Here, a is the price per user and M is the total market size. Chalamandaris and Malliaris later expanded on this work by using Itô's Calculus to bridge the gap between predictable growth and market randomness [3]. They defined the Stochastic Differential Equation as:

$$dR(t) = \mu R_t dt + \sigma R_t dW_t$$

With this, σ represents market volatility and dW_t represents random shocks. They also introduced a time-varying drift, which allows the expected growth rate of the model to change dynamically over time. These components are combined into a single simulation equation, R.

$$\mu(t) = \frac{1}{F(t)} \cdot \frac{dF(t)}{dt} \qquad R_{t+\Delta t} = R_t \cdot e^{((\mu_t - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t}Z)}$$

Where $Z \sim \mathcal{N}(0, 1)$. This enables projection of multiple potential revenue paths, accounting for both structured growth and random market volatility.

4. Survival Analysis. Some studies use variations of the Cox proportional hazards model to determine the probability of survival, which depends on a weighted mix of 7 factors: financing, collaboration, intellectual property, human capital, founder demographics, industry, and interactions [5].

$$\text{Survival}(t) = \alpha_s(t) \exp\left(\sum \beta_{s,i} F_i + \cdots + \sum \sigma_{s,i} I_i\right) + \varepsilon_s$$

More studies build on this framework by introducing more parameters, such as stratifying experience [4]. Others add more emphasis on financing's role in enabling survival [2].

5. Our Project. Building on this literature, our project aims to develop a layered 2-part simulator for startup outcomes. We plan to combine original revenue models, such as Brownian motion models of total growth or more granular models of customer adoption dynamics, with parametric survival models that capture a startup's risk of failing. In the end, we hope to achieve a realistic model of company trajectories that can be used to better inform VC decisions. We plan to tweak the parameters of our simulator using real-world data, and we hope to create a unified simulator that bridges the gap between early-stage and late-stage companies, directly showing the uncertainty inherent at each stage in our comprehensive output.

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